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Original Research Article

Pancreatic exocrine function and morphology following prior malnutrition in Asian and African cohorts: a multi-cohort study

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ABSTRACT

Background: Studies of survivors of large 20th-century famines suggest persistent after effects on metabolism and nutrient handling, including abnormal endocrine and exocrine pancreatic functions, which play key roles in nutrient digestion and absorption.

Objectives: We tested the hypothesis that severe wasting malnutrition in childhood or adulthood leads to long-term exocrine pancreatic insufficiency and/or structural abnormalities.

Methods: We followed up participants from 6 cohorts of previously malnourished (PM) and not previously malnourished (NPM) adults and children in India, the Philippines, Tanzania, and Zambia. Fecal elastase (FE-1), circulating trypsinogen, amylase, and lipase assays were performed. Ultrasound scans (USS) were conducted in all cohorts, and a subset of adults underwent computed tomography scan (CT) to estimate pancreatic volume. Statistical analyses included Kruskal–Wallis tests and logistic regression to estimate odds ratios (ORs) and 95% confidence intervals (CIs).

Results: A total of 2165 participants had assessments of pancreatic structure or exocrine function: 1799 (83%) had FE-1 measured, 1761 (81%) underwent USS, and 329 (15%) had CT. Low FE-1 (<200 μg/g) was found in 252 (32%) of 791 PM and 346 (34%) of 1008 NPM participants (OR: 0.89; 95% CI: 0.73, 1.10; $P = 0.29$). Radiological measurements of the pancreas were consistently higher in PM than NPM adult participants when adjusted for body weight; this association was reversed in children. Pancreatic volume was apparently higher, and FE-1 lower, in African compared with Asian

Abbreviations: AP, anterior–posterior diameter; CC, cranial-caudal diameter; CI, confidence interval; CICADA, chronic infections, comorbidities and diabetes in Africa, Tanzania; CLHNS, Cebu Longitudinal Health and Nutrition Survey, Philippines; CT, computed tomography scan; DICOM, Digital Imaging and Communications in Medicine; DIVIDS, Delhi Infant Vitamin D Supplementation Trial, India; EPI, exocrine pancreatic insufficiency; FE-1, fecal elastase-1; NPM, not previously malnourished; NUSTART, Nutritional Support for African Adults Starting Anti-Retroviral Therapy, Zambia; OR, odds ratio; PM, previously malnourished; PV, pancreas volume; SAM, severe acute malnutrition, Zambia; SAMPA, severe acute malnutrition in children and adults – the role of the pancreas; St-ATT, starting anti-TB treatment, Philippines; TV, transverse diameter; USS, ultrasound scan; WLZ, weight-for-length *z* score.

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cohorts, which was not due to inflammation or human immunodeficiency virus infection. Circulating trypsinogen was associated with pancreatic size only after adjusting for body weight.

Conclusions: In this large multicohort study, prior severe malnutrition did not result in exocrine pancreatic dysfunction later in life, although pancreatic size was increased in adults, reduced in children, and varied between cohorts. These findings highlight the need for further research into the long-term metabolic and nutritional consequences of early-life malnutrition across diverse populations.

Keywords: pancreatic atrophy, elastase, trypsinogen, gastrointestinal ultrasound, CT

Introduction

Famine, starvation, and undernutrition have been part of the human condition since time immemorial. Recent studies of the long-term consequences of the great famines of the 20th century have shed light on the metabolic consequences of severe undernutrition early in life. Survivors of the Dutch famine (1944–1945) are at increased risk of type 2 diabetes mellitus [1]. A systematic review provided evidence for disturbed insulin and glucose metabolism many years after recovery from severe malnutrition [2]. Pancreatic atrophy is found postmortem in children dying of severe acute malnutrition (SAM) [3,4].

Exocrine pancreatic insufficiency (EPI) signifies inadequate pancreatic enzyme production and secretion, commonly diagnosed using fecal elastase-1 (FE-1), a stable noninvasive marker [5]. Although typically associated with chronic pancreatitis, EPI is also prevalent in endocrine disorders such as type 1 and 2 diabetes, HIV infection, and childhood SAM [6–8]. Childhood SAM has been associated with possible long-term pancreatic dysfunction in survivors [9]. An association between EPI and malnutrition has been observed, although it remains unclear whether this reflects a causal relationship or mere correlation [10]. Older studies in the Ivory Coast reported that individuals with a history of malnutrition had significantly reduced pancreatic enzyme secretion compared with well-nourished controls, suggesting that early nutritional status may have lasting effects on exocrine pancreatic function [11].

Trypsinogen, amylase, and lipase are key exocrine pancreatic enzymes involved in the digestion of proteins, carbohydrates, and fats, respectively [12]. Although raised amylase and lipase concentrations in blood are primarily used to diagnose acute pancreatic inflammation, emerging evidence suggests that low circulating trypsinogen may be more indicative of chronic exocrine insufficiency [13]. Serum trypsinogen correlates significantly with pancreatic volume when normalized for BMI [14]. Pancreas volume correlates with acinar cell mass [15]. Imaging studies of the pancreas may reveal otherwise unrecognized structural pathology, including parenchymal or ductal abnormalities [16]. Tropical calcific pancreatitis has been associated with being underweight in cross-sectional studies in India, but there is also evidence of a genetic etiology [17,18]. However, there is a paucity of studies correlating pancreatic size and structural imaging abnormalities with malnutrition.

The Severe Acute Malnutrition in children and adults—the role of the Pancreas (SAMPA) study was established to investigate long-term outcomes in individuals with previously precisely documented periods of wasting malnutrition at various stages of the life course [14]. The study found glucose intolerance in survivors who became overweight or remained underweight [19]. We hypothesized that prior wasting is associated with later abnormal pancreatic structure and impaired exocrine function. Here, we report radiological measurements of pancreatic morphology, alongside measurements of FE-1; trypsinogen was measured as a possible measure of pancreatic atrophy, whereas amylase and lipase were measured to search for evidence of pancreatitis.

Methods

Study design

The SAMPA study protocol has been published and registered with clinicaltrials.gov (NCT05361148) [20]. Between October 2021 and August 2023, a total of 2275 participants were recruited into this cross-sectional re-evaluation of people followed up between 3 and 37 y after their participation in 1 of 6 previously published cohort studies. These included 2 child cohorts; Delhi Infant Vitamin D Supplementation Trial, India (DIVIDS), and SAM, Zambia and 4 adult cohorts; Chronic Infections, Comorbidities, and diabetes in Africa, Tanzania (CICADA), Nutritional Support for African Adults Starting Antiretroviral Therapy, Zambia (NUSTART), Starting Anti-TB Treatment, Philippines (St-ATT), and Cebu Longitudinal Health and Nutrition Survey, Philippines (CLHNS), although in the CLHNS study, exposure [in the previously malnourished (PM) group] occurred during childhood. Detailed information on the history and design of the previous cohorts has been published elsewhere, and a summary is provided in Supplemental Figure 1 [19,20]. All participants were requested to undergo ultrasound scans (USS) and provide stool samples. After excluding 110 (4.7%) individuals missing both investigations, analyses included 2165 participants. Participants were categorized as PM or not previously malnourished (NPM) based on cohort-specific criteria: low birth weight (<2.5 kg at term) in DIVIDS; prior hospitalization for SAM before age 2 in the SAM cohort; BMI <18.5 kg/m^2 in adults from CICADA, NUSTART, and St-ATT (linked to HIV/tuberculosis infection); and weight-for-length *z* score (WLZ) <–3 before age 2 in CLHNS, with NPM defined as always having WLZ >–2 (Table 1).

Participant Enrollment and Procedures

Participants or their guardians were contacted using recent follow-up details and invited to join the SAMPA study. Those who consented attended research clinics for enrollment, questionnaires, physical examination, and weight measurement. Stool samples were collected for FE-1, and an oral glucose tolerance test was performed [14]. Pancreatic size and structure were assessed by USS, with a randomly selected subset also undergoing computed tomography (CT). In each adult cohort, we aimed to recruit 100 participants, divided approximately evenly between PM and NPM. For African cohorts, HIV status was ascertained in the original study [19], and any testing negative previously were retested.

Exocrine pancreatic function

Exocrine pancreatic function was assessed by measuring fecal FE-1 using the ScheBo Pancreatic Elastase 1 fecal ELISA kit (Biotech) according to the manufacturer's instructions. Sample handling and

TABLE 1
Demographic, clinical and exposure characteristics of the SAMPA study cohorts[1]

Cohort	Exposure during childhood						Exposure as adults					
	DIVIDS (India)		SAM (Zambia)		CLHNS (Philippines)		CICADA (Tanzania)		NUSTART (Zambia)		St-ATT (Philippines)	
Status	NPM	PM	NPM	PM	NPM	PM	NPM	PM	NPM	PM	NPM	PM
N of participants	54	253	46	58	196	106	592	317	87	195	150	111
Mean age (SD), y	11.8 (1.4)	13.9 (0.8)	13.2 (2.0)	12.2 (2.0)	39.0 (0.4)	38.8 (0.5)	47.3 (11.0)	49.3 (10.6)	40.5 (12.5)	44.9 (7.9)	52.2 (13.7)	47.8 (16.4)
Female	25 (46%)	125 (49%)	26 (57%)	30 (52%)	72 (37%)	38 (36%)	358 (60%)	164 (52%)	75 (86%)	110 (56%)	49 (33%)	33 (30%)
HIV positive[2]	NT	NT	0 (0%)	2 (3%)	NT	NT	306/572 (54%)	253/312 (81%)	39/86 (45%)	195 (100%)	NT	NT
Mean weight (SD), kg	35.3 (13.1)	40.7 (9.3)	39.9 (11.6)	35.2 (7.3)	72.3 (14.4)	63.6 (13.0)	66.2 (13.6)	54.5 (8.9)	68.6 (17.3)	55.0 (9.9)	57.4 (9.9)	45.6 (7.6)
Median BMI,[3] (IQR)	15 (14,18)	17 (15, 19)	17 (15, 19)	16 (15, 18)	27 (25, 31)	24 (22, 28)	24 (21, 27)	19 (18, 22)	24 (21, 28)	19 (18, 22)	22 (20, 25)	18 (17, 20)
Median MUAC (IQR)	19 (17, 22)	21 (19, 23)	22 (20, 24)	21 (19, 23)	31 (29, 34)	30 (28, 33)	30 (28, 33)	26 (24, 28)	30 (27, 33)	26 (24, 28)	28 (26, 30)	24 (22, 26)
No diabetes	53 (98%)	245 (97%)	45 (100%)	57 (98%)	108 (55%)	56 (53%)	444 (75%)	235 (74%)	72 (83%)	181 (93%)	50 (33%)	55 (50%)
Prediabetes	1 (2%)	7 (3%)	0 (0%)	1 (2%)	67 (34%)	39 (37%)	127 (22%)	76 (24%)	10 (11%)	12 (6%)	45 (30%)	35 (32%)
Diabetes	0 (0%)	0 (0%)	0 (0%)	0 (0%)	21 (11%)	11 (10%)	20 (3%)	6 (2%)	5 (6%)	2 (1%)	55 (37%)	20 (18%)
EPI positive[4]	2/21 (10%)	16/97 (16%)	21/38 (55%)	33/45 (73%)	22/188 (12%)	16/103 (16%)	248/543 (46%)	114/277 (41%)	18/75 (24%)	59/165 (36%)	35/143 (24%)	14/104 (13%)
USS	54 (100%)	251 (99%)	37 (80%)	46 (79%)	186 (95%)	104 (98%)	339 (58%)	211 (67%)	85 (98%)	188 (96%)	149 (99%)	111 (100%)
CT	0	0	0	0	50 (26%)	50 (47%)	28 (5%)	32 (10%)	28 (32%)	41 (21%)	63 (42%)	37 (33%)

Abbreviations: CICADA, chronic infections, comorbidities and diabetes in Africa, Tanzania; CLHNS, Cebu Longitudinal Health and Nutrition Survey, Philippines; CT, computed tomography scan; DIVIDS, Delhi Infant Vitamin D Supplementation Trial, India; EPI, exocrine pancreatic insufficiency; MUAC, mid-upper arm circumference; NPM, not previously malnourished; NUSTART, Nutritional Support for African Adults Starting Anti-Retroviral Therapy, Zambia; PM, previously malnourished; SAM, severe acute malnutrition, Zambia; St-ATT, starting anti-TB treatment, Philippines; USS, ultrasound scan.

[1] Where the denominator is less than the number *N* shown at the head of each column, the denominator is shown. Diabetes categories were determined by plasma glucose at 120 min during an oral glucose tolerance test: <7.8 mmol/L = normal glucose tolerance, ≥7.8 mmol/L to <11.1 mmol/L = prediabetes, ≥11.1 mmol/L = diabetes. Missing data: glucose at 120 min — 1 PM of DIVIDS, 1 NPM of SAM, 1 NPM of CICADA, 1 PM of St-ATT.

[2] NT=not tested or status unknown for DIVIDS, St-ATT, and CLHNS; HIV status was tested for all African participants.

[3] BMI was expected to be lower in the child cohorts than the adult cohorts.

[4] EPI defined as elastase <200 μg/g stool.

analysis for FE-1 were conducted according to the same standard operating procedures in all study laboratories to minimize assay variability. Stool consistency was assessed at the time of collection, and visibly watery or diarrheal stool samples were excluded from analysis to minimize dilution-related false low FE-1 results. Only formed or semiformed stool samples were processed. After collection, stool samples were aliquoted and stored frozen at –20°C until analysis. The ScheBo Master Quick-Prep kit was utilized for standardized stool collection, employing its in-built dosing loop to ensure uniform sample measurements across all the study sites. ELISA methods were standardized across sites for quality control, with high values truncated at 600 μg/g. Initial plates were run in duplicate; subsequent samples were run singly because of low variability. Each plate included ≥1 control provided by the manufacturer, and interassay coefficient of variation was maintained <10.6%. EPI was defined as FE-1 <200 μg/g stool and further categorized as mild to moderate (100–199 μg/g) or severe (<100 μg/g) categories. Pancreatic amylase was measured using a Cobas Analyzer c502 (Roche Diagnostics) on 1900 available serum samples. Serum lipase was measured using a Cobas Pro Analyzer (Roche Diagnostics) on 322 DIVIDS samples. Serum trypsinogen was measured using the DELFIA Neonatal IRT kit (Revvity) with fluorescence detection performed on a Hidex Plate Reader (Hidex Oy) on all the samples from the CT subgroup. High amylase was defined as >2× the upper reference limit based on age (1–10 y: 62U/L; 10–18 y: 78 U/L age; >18 y: 106 U/L) and low amylase as <13 U/L [21].] High lipase was defined as >160U/L. For trypsinogen, levels >60 ng/mL were considered high and <20 ng/mL as low.

Pancreatic size and structure

All consenting participants underwent USS, whereas a randomly selected subgroup of adults also had CT, which was conducted without contrast to avoid related side effects given that these were volunteers. Females of childbearing age underwent pregnancy testing; children and pregnant females were excluded from CT because of radiation risks. Imaging of the pancreas was conducted after ≥6 h of fasting, using radiological equipment listed in Supplemental Table 1. Standardization of imaging measurement techniques to reduce intra-observer and interobserver variation across the sites is detailed in Supplemental Tables 2 and 3.

Transabdominal USS examinations were conducted using a 3.5-MHz curvilinear probe for all participants. Pancreas head size was measured in anterior–posterior diameter (AP) and transverse diameter (TV), whereas pancreas body and tail sizes were measured in the AP view. Additionally, in the CT subgroup, pancreas body length and tail were measured in AP views, whereas cranial-caudal diameter

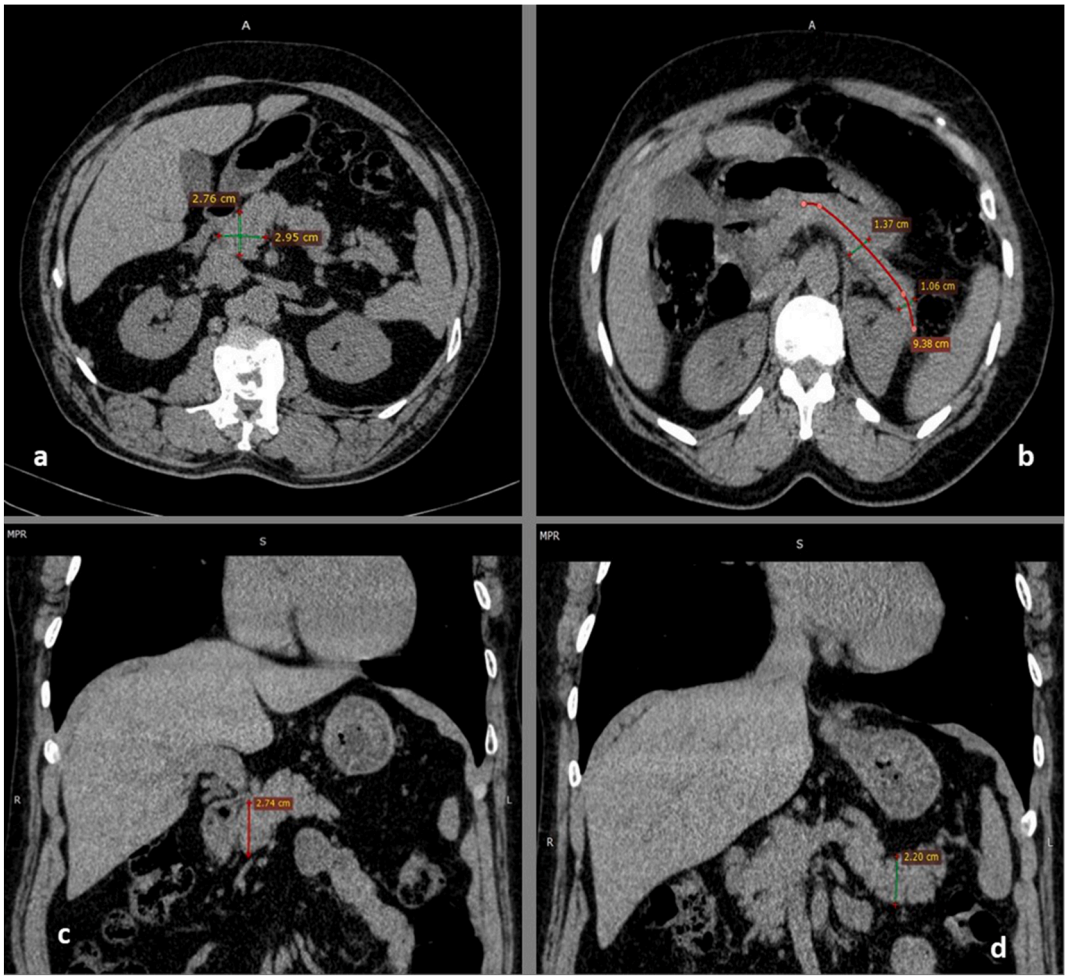


FIGURE 1. CT images of the upper abdomen showing how pancreatic axial (A) anterior–posterior and transverse head diameters; (B) anterior–posterior body and tail anterior–posterior diameters as well as neck-to-tail length; pancreatic coronal cranio-caudal (C) head and (D) body diameters were measured. CT, computed tomography.

(CC) of the pancreas body and tail were also recorded (Figure 1). A CT-derived pancreas volume (PV) was calculated using a validated formula [22]:

$$PV = [(AP_{tail}+AP_{body})/2\times L_{body\&tail}\times CC_{body}]+[(AP_{head}/2)^2\times 3.14\times CC_{head}]$$

where AP_{tail}= pancreas tail diameter, AP_{body}= pancreas body AP diameter, $L_{body\&tail}$= pancreas body and tail length, CC_{body} = pancreas body diameter, AP_{head} =pancreas head AP diameter, and CC_{head}= pancreas head CC diameter. The shape, contour, parenchymal or ductal configuration, presence of pancreatic calcifications, and any other abnormalities were documented. Imaging data, fully anonymized with no patient identifiers, were uploaded for review and archiving to secure servers on RedCAP version 14.1.2 (Vanderbilt University) for USS and PostDICOM (PostDICOM B.V.) for CT images. All USS and CT images followed a standardized image acquisition protocol in all cohorts, which included being verified for quality by an experienced onsite radiologist blinded to the malnutrition status of the participants.

Data analysis

Data were checked for missing and outlying values and analyzed using Stata 18 (StataCorp LLC) and R (https://www.r-project.org/). The primary comparison was between PM and NPM participants. Participant characteristics, and laboratory and radiological measurements were represented using counts and percentages, or medians and IQRs because of non-normal distributions. Any missing data were excluded from analysis and declared in Tables 1 and 2. The Kruskal–Wallis test was used for PM and NPM group comparisons, with additional analyses exploring associations of cohort, age, sex, and HIV using Kruskal–Wallis and/or Spearman's correlation. Logistic regression was used to determine multivariate associations with categorical EPI, whereas linear regression was used for associations with continuous outcomes selected on the basis of hypotheses about potential determinants. To adjust pancreatic measurements for body size, we also presented these measurements as a ratio (divided by) body weight (kg).

Ethical considerations

Ethical approval was obtained from the ethics committees of the London School of Hygiene and Tropical Medicine, United Kingdom (22792-2); National Institute for Medical Research, Tanzania (NIMR/HQ/R.8a/Vol. IX/3686); University of Zambia Biomedical Research Ethics Committee, Zambia (1686-2021); Institute of Home Economics, University of Delhi, India (IHE/2021-22/Admin/Ethics.com/684(A)); and St. Frances Cabrini Medical Centre—Asian Eye Institute, Philippines (2021-012). Adult participants gave written informed consent. Children provided assent, and their parents or guardians gave written informed consent. Participants found with abnormal findings were referred for appropriate local healthcare management. All available data were analyzed.

TABLE 2
Radiological measurements in the whole cohort, by prior exposure[1]

Adult cohorts						Child cohorts				
Measurement (units)	NPM		PM		*P*	NPM		PM		*P*
	Diameter	*n*	Diameter	*n*		Diameter	*n*	Diameter	*n*	
USS measurements unadjusted (n = 1761 contributing any measurement)										
Head, transverse (cm)	2.13 (1.94, 2.39)	547	2.02 (1.57, 2.35)	476	<0.001	1.53 (1.33, 1.93)	88	1.58 (1.38, 1.84)	287	0.94
Head, AP (cm)	2.10 (1.88, 2.40)	625	2.10 (1.77, 2.50)	539	0.410	2.02 (1.81, 2.20)	90	2.05 (1.84, 2.27)	297	0.34
Body, AP (cm)	1.26 (1.10, 1.50)	655	1.21 (1.00, 1.49)	567	0.002	1.29 (1.05, 1.46)	90	1.29 (1.11, 1.5)	295	0.40
Tail, AP (cm)	1.23 (1.07, 1.60)	421	1.46 (1.17, 1.84)	402	<0.001	1.48 (1.22, 1.80)	88	1.45 (1.27, 1.66)	289	0.51
USS measurements adjusted for body weight[2]										
Head, transverse (cm/kg)	0.033 (0.027, 0.038)	547	0.036 (0.028, 0.044)	475	<0.001	0.047 (0.035, 0.058)	88	0.040 (0.033, 0.050)	283	0.002
Head, AP (cm/kg)	0.034 (0.029, 0.039)	625	0.040 (0.032, 0.047)	538	<0.001	0.060 (0.043, 0.072)	90	0.051 (0.043, 0.060)	293	0.003
Body, AP (cm/kg)	0.019 (0.017, 0.024)	655	0.022 (0.018, 0.028)	566	<0.001	0.037 (0.027, 0.049)	90	0.033 (0.027, 0.041)	291	0.047
Tail, AP (cm/kg)	0.019 (0.016, 0.025)	421	0.027 (0.019, 0.035)	401	<0.001	0.044 (0.035, 0.052)	88	0.037 (0.031, 0.044)	285	<0.001
CT measurements unadjusted (n = 329 contributing any measurement)[3]										
Head, transverse (cm)	2.64 (2.41, 2.94)	168	2.66 (2.35, 3.00)	156	0.77	—	—	—	—	—
Head, AP (cm)	2.58 (2.30, 2.83)	169	2.50 (2.20, 2.80)	158	0.14	—	—	—	—	—
Body, AP (cm)	2.07 (1.72, 2.36)	168	2.12 (1.76, 2.44)	158	0.16	—	—	—	—	—
Tail, AP (cm)	1.76 (1.44, 2.00)	169	1.78 (1.51, 2.10)	156	0.40	—	—	—	—	—
Length (cm)	10.43 (9.35, 11.42)	122	9.83 (8.95, 10.84)	129	0.005	—	—	—	—	—
Pancreas volume (cm^3)	64.94 (52.17, 81.10)	119	62.12 (48.44, 81.58)	123	0.69	—	—	—	—	—
CT measurements adjusted for body weight[2]										
Head, transverse (cm/kg)	0.042 (0.036, 0.049)	168	0.049 (0.039, 0.060)	156	<0.001	—	—	—	—	—
Head, AP (cm/kg)	0.041 (0.035, 0.048)	169	0.045 (0.039, 0.054)	158	<0.001	—	—	—	—	—
Body, AP (cm/kg)	0.032 (0.026, 0.038)	168	0.037 (0.031, 0.047)	158	<0.001	—	—	—	—	—
Tail, AP (cm/kg)	0.028 (0.023, 0.034)	169	0.032 (0.027, 0.041)	156	<0.001	—	—	—	—	—
Length (cm/kg)	0.161 (0.140, 0.181)	122	0.172 (0.148, 0.198)	129	0.006	—	—	—	—	—
Pancreas volume (cm^3/kg)	1.02 (0.83, 1.21)	119	1.12 (0.87, 1.49)	123	0.01	—	—	—	—	—

Abbreviations: AP, anterior–posterior diameter; CT, computed tomography scan; NPM, not previously malnourished; PM, previously malnourished; USS, ultrasound scan.

[1] All unadjusted measurements shown are median (IQR), in cm, except for pancreatic volume in cm^3. Adjusted measurements are median (IQR) of the ratio of the measurement (cm or cm^3) to body weight (kg).

[2] Weight missing for 1 adult and 4 children.

[3] CT was not done in the child cohorts.

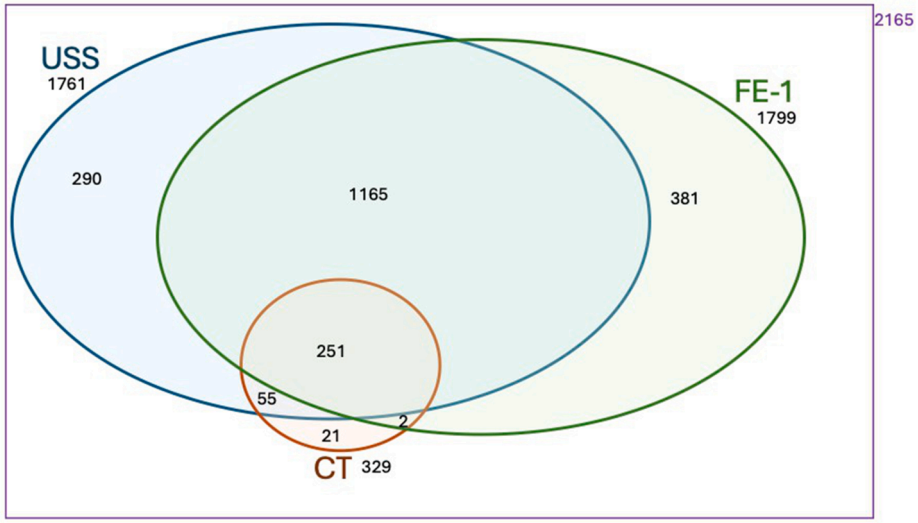


FIGURE 2. Venn diagram showing an overlap in participants with FE-1, USS, and CT measures. CT, computed tomography; FE-1, fecal elastase; USS, ultrasound.

Results

Study participants in 6 cohorts

The entire SAMPA study cohort included 2275 individuals (Supplementary Figure 1). However, 110 did not attend any of the procedures, leaving a total of 2165 participants included from the 6 study cohorts; this included 411 children and 1754 adults (Table 1). Of these 2165 participants, 1799 had FE-1 measured, 1761 were investigated by USS, and 329 by CT (Figure 2).

Exocrine function and prior malnutrition

Low FE-1, indicating EPI, was found in 598 (33%) of 1799 individuals, and very low FE-1, indicating severe EPI, in 324 (18%). Low FE-1 was found in 252 (32%) of 791 PM participants and 346 (34%) of 1008 NPM participants [odds ratio (OR): 0.89; 95% confidence interval (CI): 0.73, 1.10; $P = 0.29$]. There was no association between PM and FE-1 concentration in individual study cohorts (Figure 3). FE-1 was lower in African participants (median 237 µg/g; IQR: 103–487) than in Asian participants (median 566 µg/g; IQR: 297–600; $P < 0.001$). EPI was also more frequent in African than Asian participants (OR: 3.98, 95% CI: 3.12, 5.10; $P < 0.0001$). Similarly, the prevalence of severe EPI was higher in Africans (OR: 4.89; 95% CI: 3.51, 6.94; $P < 0.0001$). Pooling all cohorts, logistic regression was used to model EPI, including childhood exposure (the 2 child cohorts plus CLHNS), age, sex, and African/Asian residence. In the final model, only African residence was associated with EPI (OR: 3.98, 95% CI: 3.1, 5.1; $P < 0.001$). In African cohorts, where HIV status was measured, HIV infection was associated with higher FE-1 levels (Supplemental Figure 2), and so HIV did not explain lower FE-1 levels in Africa. As HIV infection was measured almost exclusively in Africa, both HIV and Africa could not be included in regression models due to collinearity.

Correlation between USS and CT measurements

USS and CT pairwise pancreatic measurements of the head and body showed weak correlations whereas the tail showed very weak correlations (Supplemental Figure 3). Unlike CT, this is likely to be related to technical challenges in USS related to bowel gas, especially over the body and tail, causing poor image resolution and impeding clear delineation of pancreatic margins. This suggests that USS has limited reliability in pancreatic measurements compared with CT.

Pancreatic size, structure, and prior malnutrition

When adjusted for body weight, pancreatic dimensions were consistently slightly larger in adults with PM than NPM individuals (Table 2), but this varied in individual cohorts (Figure 4). In children, USS pancreatic measurements in the PM group were smaller than in NPM (Figure 5). Pancreatic USS dimensions were apparently greater in African adult cohorts than Asian adult cohorts, but this difference was not apparent in children (Supplemental Table 4). Structural abnormalities were minimal, seen in 7.6% of NPM compared with 4.0% PM participants on USS with 6.0% of NPM compared with 2.9% PM participants presenting with parenchymal steatosis (Supplemental Tables 5 and 6 and B).

Using linear regression modeling of square root transformed pancreatic volume (measured by CT, adjusted for weight as cm^3/kg), including PM/NPM, sex, child cohort, childhood exposure, Africa/Asia, low elastase, and high amylase, only Africa was significantly and positively associated with CT-derived volume ($\beta = 0.099$; 95% CI: 0.053, 0.145; $P < 0.001$). Using similar linear regression of log-transformed USS measurements, child cohort, African cohort, and serum amylase were positively associated with all measurements, and PM was positively associated with 3 of 4 measures (Supplemental Table 7). Sex was not associated with pancreatic volume or USS measurements in any model. To confirm consistency, when individual cohorts were analyzed instead of Africa/Asia as a binary variable, the same pattern was evident, with CLHNS and St-ATT cohorts consistently showing lower pancreatic volume.

EPI and pancreatic size

To determine whether pancreas size correlates with EPI, we compared pancreatic dimensions in participants with or without EPI. Using USS evaluation, 3 parameters (pancreas head AP, body, and tail diameters) were greater in participants with EPI than those without (Supplemental Figure 4). Neither the pancreas head transverse diameter nor any parameters by CT differed in EPI (not shown).

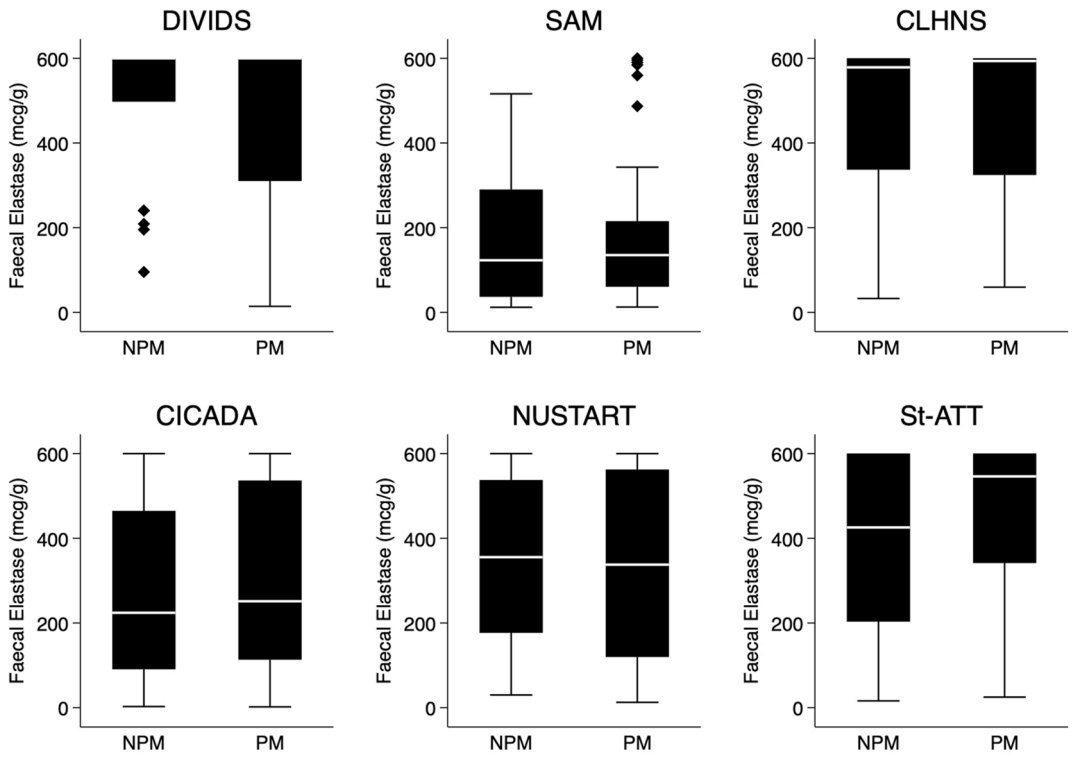


FIGURE 3. Fecal elastase levels by malnutrition status in each cohort. A value of <200 μg/g was regarded as evidence of exocrine pancreatic insufficiency. Values were truncated at 600 μg/g. CICADA, chronic infections, comorbidities and diabetes in Africa, Tanzania; CLHNS, Cebu Longitudinal Health and Nutrition Survey, Philippines; DIVIDS, Delhi Infant Vitamin D Supplementation Trial, India; NPM, not previously malnourished; NUSTART, Nutritional Support for African Adults Starting Anti-Retroviral Therapy, Zambia; μg/g, microgram per gram of stool; PM, previously malnourished; SAM, severe acute malnutrition; St-ATT, starting anti-TB treatment, Philippines.

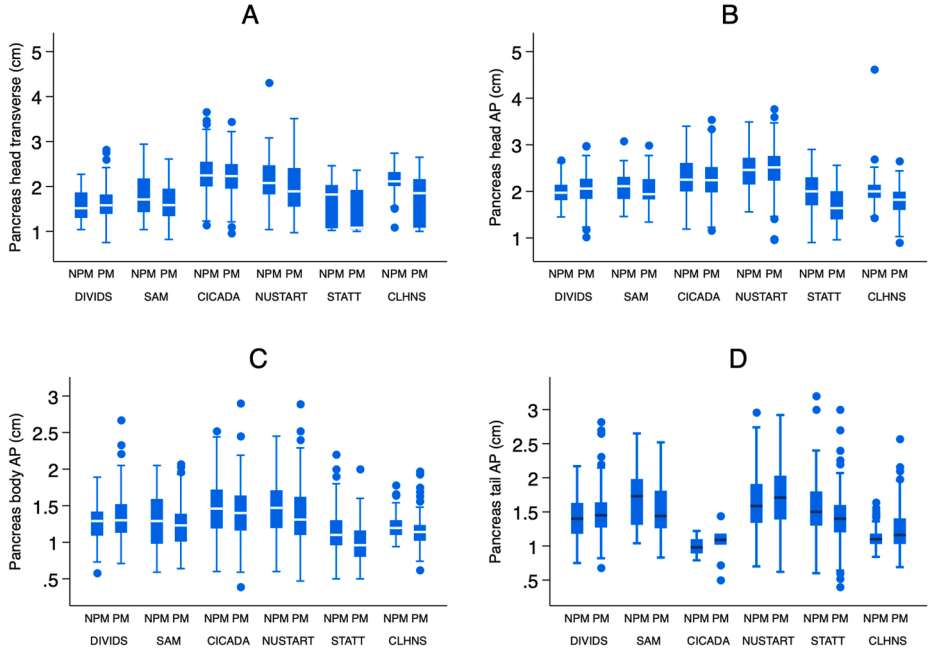


FIGURE 4. USS pancreatic measurements in relation to malnutrition status by cohort. CICADA, chronic infections, comorbidities and diabetes in Africa, Tanzania; AP, anterior–posterior diameter; CLHNS, Cebu Longitudinal Health and Nutrition Survey, Philippines; DIVIDS, Delhi Infant Vitamin D Supplementation Trial, India; NPM, not previously malnourished; NUSTART, Nutritional Support for African Adults Starting Anti-Retroviral Therapy, Zambia; OAP, anterior–posterior diameter; PM, previously malnourished; SAM, severe acute malnutrition; St-ATT, starting anti-TB treatment, Philippines; USS, ultrasound scan.

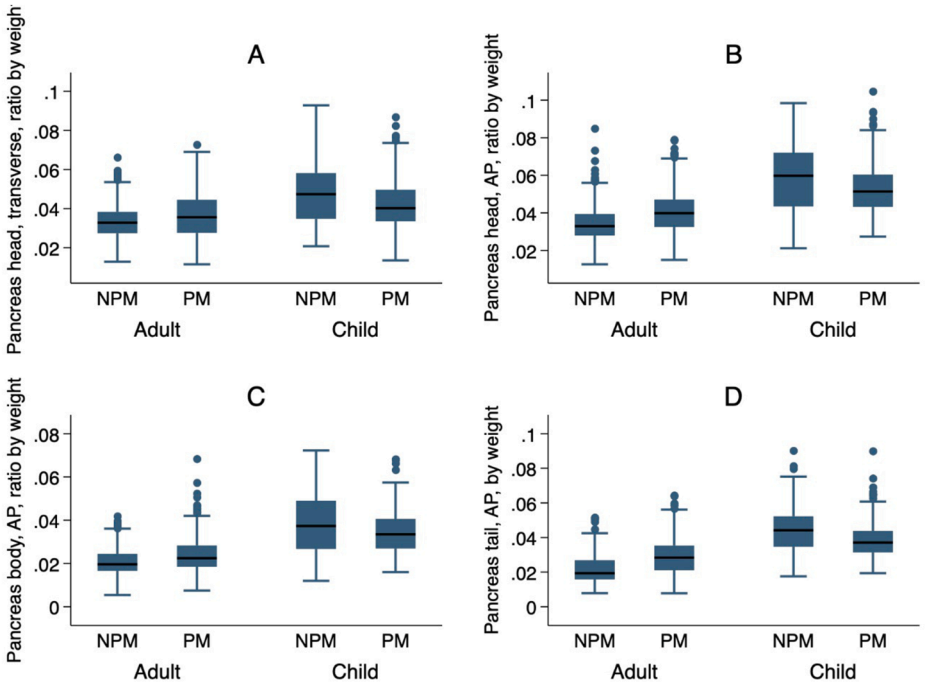


FIGURE 5. Relationship of USS pancreatic dimensions adjusted for body weight, as a ratio, in adults and child cohorts. (A) Pancreatic head transverse; (B) pancreatic head AP; (C) pancreatic body AP; and (D) pancreatic tail AP diameters. Pancreatic dimensions were higher in the PM than the NPM group ($P = 0.0001$ for all comparisons in adults, whereas for children $P = 0.047$ for body, but $P < 0.003$ for all other measures). Adult cohorts included NUSTART, CICADA, St-ATT, and CLHNS, whereas child cohorts included SAM and DIVIDS. AP, anterior–posterior diameter; CLHNS, Cebu Longitudinal Health and Nutrition Survey, Philippines; DIVIDS, Delhi Infant Vitamin D Supplementation Trial, India; NPM, not previously malnourished; NUSTART, Nutritional Support for African Adults Starting Anti-Retroviral Therapy, Zambia; PM, previously malnourished; SAM, severe acute malnutrition; St-ATT, starting anti-TB treatment, Philippines.

Serum biomarkers of inflammation and atrophy

High amylase was observed in 3 of 1900 measurements, high lipase in 0 of 322 measurements, and high trypsinogen in 13 of 300 measurements. Amylase concentrations were higher in the PM group (Supplemental Table 8). Low trypsinogen was found in 102 (34%) of samples tested; it was not associated with PM (OR: 2.2; 95% CI: 0.60, 10.0; $P = 0.26$), nor with EPI (OR: 1.1; 95% CI: 0.56, 2.1; $P = 0.87$). Trypsinogen was weakly associated with pancreatic dimensions, but much more consistently when radiological measurements were adjusted for body weight (Supplemental Figure 5).

Discussion

Our hypothesis that PM would lead to a smaller pancreas size and poorer exocrine function when measured years later, was not supported by our results. Indeed, radiological data from 1761 USS and 329 CT showed evidence of marginally greater size in PM individuals. This increase was much more apparent after adjusting pancreatic size for body weight because, overall, PM participants had a lower weight than NPM at the time of pancreas assessment. We investigated whether this increase might be attributable to inflammation, but this was not apparent radiologically. There was statistically significant evidence of increased inflammation using amylase, but the absolute increase was minimal. There was more evidence of inflammation using serum trypsinogen, but the highest pancreatic volumes were not accompanied by high circulating trypsinogen concentrations, so overall, there was no evidence that this unexpected finding is due to inflammation [23]. The SAMPA study has found evidence that PM and obesity interact in contributing to diabetes risk, so there may be an effect on endocrine pancreatic function [19]. Taken together, these data do not support the hypothesis that any impact of malnutrition on endocrine pancreatic function is a consequence of pancreatic atrophy. We found very infrequent pancreatic calcification, irrespective of prior malnutrition. This contrasts with evidence, largely from the Indian subcontinent, of tropical calcific pancreatic disease following malnutrition [2,17,18]. This would be rare in the age group included in the DIVIDS cohort, and our analysis did not include Indian adults, so it is possible that this syndrome may not have been captured in the SAMPA study.

The high prevalence of EPI is surprising, at 33%, in both African (Tanzania, Zambia) and Asian (India, Philippines) cohorts. In our study, HIV-positive participants had higher FE-1 than HIV-negative, so their inclusion acts to attenuate the already observed higher odds of low FE-1 in African cohorts and not inflate it. This means the Africa–Asia difference in EPI prevalence reported is a conservative estimate, and removing the HIV-positive participants would further make the African FE-1 even lower, and the Africa–Asia OR even larger. This effectively negates HIV as an explanation for the low African FE-1 and further strengthens our findings. Putting aside any stool or assay measurement variations, another aspect to consider is whether the FE-1 <200 μg/g cut off we used truly reflects EPI in asymptomatic heterogeneous cohorts, because we did not include

supportive tests such as 72-h fecal fat or secretin pancreatic function in this study. As neither PM nor HIV can explain this, we are left with no clear explanation. The prevalence of low circulating trypsinogen levels was also high. Our data provide partial support for the idea that low trypsinogen is a marker of a smaller pancreas, especially when pancreatic size is adjusted for body weight [14]. Although serum amylase and lipase are not recommended diagnostic tests for exocrine pancreatic insufficiency, their inclusion was primarily to identify evidence of acute or chronic pancreatic inflammation that might explain differences in pancreatic morphology. Trypsinogen was included because low concentrations have been previously shown to reflect reduced acinar cell mass and pancreatic reserve rather than transient changes in secretory activity and may correlate with pancreatic volume.

Previous work on famine survivors suggests metabolic consequences of PM, notably increased incidence of impaired glucose tolerance in later life [2]. The apparent consistent lack of major effects on pancreatic size or exocrine function in the SAMPA study suggests that this is an effect likely to be limited to the pancreatic islets, where pancreatic hormones are secreted, or effects on insulin tolerance mediated centrally or peripherally. There is existing literature suggesting that ~90%–95% of acinar cell mass must be destroyed before clinically evident malabsorption occurs, which can often lead to the dissociation between modest reductions in pancreatic volume and absence of correlation with EPI [24]. Furthermore, functional output may depend not only on acinar mass but also on acinar integrity, neural and hormonal regulation, ductal physiology, and nutrient stimulation. Our findings thus reinforce the concept that structural and functional pancreatic measures should not be considered interchangeable.

Our data suggest that there is a difference in pancreatic size between African and Asian populations, with larger pancreatic dimensions in Africa, accompanied by lower FE-1. The significance, or even generalizability, of these differences is unknown. One possibility is that steatosis could explain both findings, and steatosis was apparent radiologically in the CICADA cohort. Steatosis and fibrosis have been reported histologically as distinct phenotypes in a recent postmortem study of British diabetics [25]. Older studies comparing West African and European populations have shown that individuals from Senegal and the Ivory Coast often exhibit lower markers of exocrine pancreatic function, such as FE-1, duodenal fluid volume, bicarbonate, and phospholipase levels. These differences were observed even in well-nourished individuals, highlighting the potential influence of regional and environmental factors, such as diet, when evaluating EPI in populations with a history of malnutrition [11,26]. The Africa–Asia binary classification grew out of the observation that the cohorts appear to group after continent. We performed regression with both individual cohorts included and continents included and obtained similar results using both approaches. Further work is needed to confirm and/or explain this observation.

The strength of the study lies in its size and in the careful ascertainment of exposure, between 3 and 37 y previously, at the inception of these various cohorts in 2 different countries per continent. We consider the heterogeneity of the cohorts and the definitions of prior malnutrition to be a strength of the SAMPA study and were an intentional part of the original design. We were interested in whether there was some underlying long-term effect of malnutrition, irrespective of the original cause of the malnutrition that may include low birth weight, SAM, tuberculosis, or HIV-related malnutrition. This study also has clear limitations. Baseline pancreas assessment was not possible within the SAMPA study, which was a cross-sectional study of pancreas after, not during, malnutrition. Therefore, a change in pancreas morphology and function over time was not the aim and, in fact, would have been difficult to interpret for cohorts in which malnutrition occurred in infancy or early childhood, but pancreas assessment was many years later. There is published literature on pancreatic impairments during malnutrition [2,20], which is why we were interested in determining whether these had any long-term consequences. To avoid the need for sedation and exposure to radiation, we did not carry out CT in children, and to reduce the discomfort of the procedure, we considered it unwise to administer intravenous contrast, which may cause unpleasant flushing sensations. Consequently, our measurement of pancreatic dimensions is based on non-contrast–enhanced radiological measurements. The most significant limitation was the difficulty in making comparisons across continents, which necessitated cross-comparisons between radiologists and between laboratories. Although every effort was made to standardize these measures, there remains the possibility that unmeasured systematic bias could have contributed to the geographical differences observed.

In conclusion, this large study estimated the long-term effects of prior malnutrition on the pancreas and found that exocrine function was unaffected, but pancreatic size was increased in adults and reduced in children.

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Author contributions

The authors' responsibilities were as follows — GP, JAS, GTK, PK, DN, JAMS, SF: designed the research; PSN, MJT, BK, PK, NRL, CC, JAS: collected and verified the data; PSN, MJT, PK, JAS: conducted the statistical analysis and drafted the initial manuscript; PSN, MJT, PK: primary responsibility for final content; and all authors: read and approved the final manuscript and reviewed and edited the manuscript.

Conflict of interest

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Data availability

Data described in the manuscript will be made available on request to the corresponding author. Sharing will usually be granted for bona fide research conducted within the terms of the original ethical approvals and involving members of the research team.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work, the authors used OpenAI's ChatGPT (GPT-4) to organize references in the format recommended by the journal. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ajcnut.2026.101476.

References

[1] C. Li, C. Ó Gráda, L.H. Lumey, Famine mortality and contributions to later-life type 2 diabetes at the population level: a synthesis of findings from Ukrainian, Dutch, and Chinese famines, BMJ Glob. Health. 9 (8) (2024) e015355, https://doi.org/10.1136/bmjgh-2024-015355.

[2] F. Ferdous, S. Filteau, N.B. Schwartz, S. Gumede-Moyo, S.E. Cox, Association of postnatal severe acute malnutrition with pancreatic exocrine and endocrine function in children and adults: a systematic review, Br. J. Nutr. 129 (4) (2023) 576–609, https://doi.org/10.1017/S0007114522001404.

[3] H.C. Trowell, J.N.P. Davies, R.F.A. Dean, Kwashiorkor, Edward Arnold, London, 1954, p. 123, 125, 146, 148.

[4] J.C. Waterlow, Protein-energy malnutrition, Arnold, London, 1992, pp. 65–66.

[5] S. Lüth, S. Teyssen, K. Forssmann, C. Kölbel, F. Krummenauer, M.V. Singer, Fecal elastase-1 determination: gold standard of indirect pancreatic function tests? Scand. J. Gastroenterol. 36 (10) (2001) 1092–1099, https://doi.org/10.1080/003655201750422729.

[6] R.H. Bartels, S.L. Meyer, T.A. Stehmann, C. Bourdon, R.H. Bandsma, W. P. Voskuijl, Both exocrine pancreatic insufficiency and signs of pancreatic inflammation are prevalent in children with complicated severe acute malnutrition: an observational study, J. Pediatr. 174 (2016) 165–170, https://doi.org/10.1016/j.jpeds.2016.04.013.

[7] M. Piciucchi, G. Capurso, L. Archibugi, M.M. Delle Fave, M. Capasso, G. Delle Fave, Exocrine pancreatic insufficiency in diabetic patients: prevalence, mechanisms, and treatment, Int. J. Endocrinol. 2015 (2015) 595649, https://doi.org/10.1155/2015/595649.

[8] A. Carroccio, L. Di Prima, C. Di Grigoli, M. Soresi, E. Farinella, D. Di Martino, et al., Exocrine pancreatic function and fat malabsorption in human immunodeficiency virus-infected patients, Scand. J. Gastroenterol. 34 (7) (1999) 729–734, https://doi.org/10.1080/003655299750025958.

[9] K. Grey, G.B. Gonzales, M. Abera, N. Lelijveld, D. Thompson, M. Berhane, et al., Severe malnutrition or famine exposure in childhood and cardiometabolic non-communicable disease later in life: a systematic review, BMJ Glob. Health. 6 (3) (2021) e003161, https://doi.org/10.1136/bmjgh-2020-003161.

[10] R.H. Bartels, D.A. van den Brink, R.H. Bandsma, M. Boele van Hensbroek, M.M. Tabbers, W.P. Voskuijl, The relation between malnutrition and the exocrine pancreas: a systematic review, J. Pediatr. Gastroenterol. Nutr. 66 (2) (2018) 193–203, https://doi.org/10.1097/MPG.0000000000001769.

[11] J.F. Sauniere, H. Sarles, Exocrine pancreatic function and protein-calorie malnutrition in Dakar and Abidjan (West Africa): silent pancreatic insufficiency, Am. J. Clin. Nutr. 48 (5) (1988) 1233–1238, https://doi.org/10.1093/ajcn/48.5.1233.

[12] M. Karpińska, M. Czauderna, Pancreas—its functions, disorders, and physiological impact on the mammals' organism, Front. Physiol. 13 (2022) 807632, https://doi.org/10.3389/fphys.2022.807632.

[13] N. Bush, V.S. Akshintala, Interpretation of serum pancreatic enzymes in pancreatic and nonpancreatic conditions, Curr. Opin. Gastroenterol. 39 (5) (2023) 403–410, https://doi.org/10.1097/MOG.0000000000000961.

[14] J.J. Ross, C.H. Wasserfall, R. Bacher, D.J. Perry, K. McGrail, A.L. Posgai, et al., Exocrine pancreatic enzymes are a serological biomarker for type 1 diabetes staging and pancreas size, Diabetes 70 (4) (2021) 944–954, https://doi.org/10.2337/db20-0995.

[15] R.R. Almeida, G.C. Lo, M. Patino, B. Bizzo, R. Canellas, D.V. Sahani, Advances in pancreatic CT imaging, AJR Am. J. Roentgenol. 211 (1) (2018) 52–66, https://doi.org/10.2214/AJR.17.18665.

[16] I.K. Nordaas, E. Tjora, G. Dimcevski, I.S. Haldorsen, S.S. Olesen, A. M. Drewes, et al., Structural imaging findings are related to clinical complications in chronic pancreatitis, United European Gastroenterol. J 10 (4) (2022) 385–395, https://doi.org/10.1002/ueg2.12228.

[17] S. Paliwal, S. Bhaskar, G.R. Chandak, Genetic and phenotypic heterogeneity in tropical calcific pancreatitis, World J. Gastroenterol. 20 (46) (2014) 17314–17323, https://doi.org/10.3748/wjg.v20.i46.17314.

[18] E. Bhatia, G. Choudhuri, S.S. Sikora, O. Landt, A. Kage, M. Becker, et al., Tropical calcific pancreatitis: strong association with SPINK1 trypsin inhibitor mutations, Gastroenterology 123 (4) (2002) 1020–1025, https://doi.org/10.1053/gast.2002.36028.

[19] D.P. Paglinawan-Modoc, S. Ahmed, M. Chisenga, S.E. Cox, R. Dasgupta, P. Duazo, et al., Associations of prior wasting malnutrition with later indicators of glucose tolerance across 4 countries in Africa and Asia: the Severe Acute Malnutrition - the Role of the Pancreas (SAMPA) study, Am. J. Clin. Nutr. 122 (4) (2025) 986–996, https://doi.org/10.1016/j.ajcnut.2025.07.032.

[20] S. Ahmed, G. PrayGod, N.R. Lee, P. Kelly, G. Trilok-Kumar, M. Chisenga, et al., Long-term health after severe acute malnutrition in children and adults—the role of the pancreas (SAMPA): protocol, F1000Res 11 (2022) 777, https://doi.org/10.12688/f1000research.123389.2.

[21] M. Jalal, S.A. Gbadegesin, N. Tehami, K. Nakajima, What is the clinical significance of low serum amylase? Systematic review of the conditions associated with low serum amylase, Frontline Gastroenterol 15 (2) (2023) 154–161, https://doi.org/10.1136/flgastro-2023-102405.

[22] A. Djuric-Stefanovic, D. Masulovic, J. Kostic, K. Randjic, D. Saranovic, CT volumetry of normal pancreas: correlation with the pancreatic diameters measurable by the cross-sectional imaging, and relationship with the gender, age, and body constitution, Surg. Radiol. Anat. 34 (9) (2012) 811–817, https://doi.org/10.1007/s00276-012-0962-7.

[23] M. Chen, H. Fang, J. Wu, Y. Huang, F. Cheng, Y. Qin, et al., Establishment of trypsinogen-2 amplification luminescent proximity homogeneous assay and its application in acute pancreatitis, J. Fluoresc. 35 (7) (2025) 5355–5366, https://doi.org/10.1007/s10895-024-03917-0.

[24] E.P. DiMagno, V.L.W. Go, W.H.J. Summerskill, Relations between pancreatic enzyme outputs and malabsorption in severe pancreatic insufficiency, N. Engl. J. Med. 288 (16) (1973) 813–815, https://doi.org/10.1056/NEJM197304192881603.

[25] N.J. Dyson, N. Kattner, Y. Al-Selwi, M. Honkanen-Scott, M.F. Shaw, C.A. Brack, et al., Quantitative analysis of human adult pancreatic histology reveals separate fatty and fibrotic phenotypes in type 2 diabetes, Diabetologia 68 (12) (2025) 2840–2853, https://doi.org/10.1007/s00125-025-06547-8.

[26] J.F. Sauniere, H. Sarles, Y. Attia, A. Lombardo, T.N. Yoman, R. Laugier, et al., Exocrine pancreatic function of children from the Ivory Coast compared to French children: effect of kwashiorkor, Dig. Dis. Sci. 31 (5) (1986) 481–486, https://doi.org/10.1007/BF01320311.